Prathamesh Nikam

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Professional Summary

M.Tech Information Technology candidate building machine learning and Generative AI systems for retrieval-augmented generation (RAG), natural language processing (NLP), recommender systems, and probabilistic forecasting. Published a Ray-based ensemble package on PyPI and deployed AI applications on Amazon Web Services (AWS) and Google Cloud Platform (GCP).

Technical Skills

Languages and Data: Python, SQL, Java, TypeScript, NumPy, Pandas, PostgreSQL
Machine Learning: PyTorch, Scikit-learn, XGBoost, Deep Learning, Ensemble Learning, Feature Engineering, Model Evaluation, Probabilistic Forecasting, Recommender Systems, Explainable AI (SHAP)
Generative AI and NLP: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Transformers, Sentence Transformers, LangChain, LangGraph, AI Agents, FAISS, ChromaDB, Whisper, spaCy, NLTK
ML Systems and Cloud: Ray, FastAPI, Streamlit, Gradio, Amazon Bedrock, AWS Lambda, GCP Cloud Run, Firebase, Cloudflare Workers, Git

Experience

University of Hyderabad — **July 2025 - June 2026**
Teaching Assistant — *Hyderabad, India*

- **IT Lab (Web Development), Jan 2026 - June 2026:** Mentored students building full-stack applications with frontend frameworks, REST APIs, databases, deployment workflows, and debugging; reviewed lab work and project implementations.
- **Object-Oriented Programming Lab (Java), July 2025 - Dec 2025:** Guided students through Java, object-oriented design, inheritance, polymorphism, exception handling, and debugging; evaluated lab work and assignments.

Projects

Shortlist'd - Evidence-First Resume Studio
Technologies: Next.js, TypeScript, Gemini, Firebase, Cloudflare Workers, LaTeX

- Built an ATS resume platform that parses PDF, DOCX, TXT, and Markdown files, compares resume content with job descriptions, compiles LaTeX in the browser, and generates tailored cover letters.
- Implemented Google authentication, UID-scoped Firestore history, guest-to-account migration, and a Cloudflare Worker proxy that forwards user-supplied Gemini keys without storing them.

ForecastForge - Probabilistic Media Decision Intelligence
Technologies: Python, XGBoost, Empirical Bayes, Monte Carlo Simulation, Streamlit

- Developed probabilistic 30/60/90-day revenue and ROAS forecasts for Google, Meta, and Microsoft Ads using empirical Bayes priors, XGBoost, cold-start peer transfer, and reconciled P10/P50/P90 intervals.
- Built leakage-safe rolling backtests and deterministic scoring; selected XGBoost after it trained 4.5x faster than CatBoost with stronger overall benchmark performance.

FH-RAG - Fuzzy Hierarchical Retrieval-Augmented Generation
Technologies: Python, PyTorch, Transformers, Sentence Transformers, Ray

- Built a hierarchical RAG pipeline for long documents using tree-structured chunking, fuzzy relevance propagation, level-aware beam search, and soft-threshold pruning.
- Evaluated six retrieval variants on SQuAD and BookSum with Recall@k, mean reciprocal rank (MRR), nDCG@5, ROUGE, F1, and faithfulness metrics.

TA-RS - Transactional Analysis Recommendation System
Technologies: Python, PyTorch, DeBERTa, BiGRU, Multi-Task Learning

- Developed a multi-task dialogue model with speaker-aware DeBERTa embeddings, causal cross-utterance attention, and a bidirectional GRU conversation encoder.
- Built weak-label and inference pipelines to jointly predict ego states, transaction types, stroke types, and speaker life positions for psychology-aware recommendation research.

HQDE - Hierarchical Quantum-Distributed Ensemble Learning
Technologies: Python, PyTorch, Ray, PyPI

- Published a PyPI package that combines quantum-inspired ensemble aggregation with Ray-based distributed training and inference.
- Implemented Byzantine fault tolerance, MapReduce weight aggregation, and adaptive quantization; evaluated the system on MNIST, CIFAR-10, and SVHN.

Projects

Neural Consensus Engine
Technologies: Python, FastAPI, Gemini, LangGraph, GCP Cloud Run
- Built a LangGraph Mixture-of-Experts workflow with creative, logical, and ethical agents plus a meta-agent that synthesizes a consensus response.
- Deployed the FastAPI application on GCP Cloud Run with configurable agent roles and live process-graph visualization.

TRACE - Transparent Results and Academic Compliance Engine
Technologies: Python, FastAPI, Scikit-learn, FAISS, ChromaDB, SHAP, Streamlit
- Built a 15-component academic ML pipeline for semantic grading, confidence scoring, anomaly detection, dropout-risk classification, and RAG-based study-resource matching.
- Integrated optical character recognition (OCR), multi-model consistency checks, SHAP explanations, and human approval thresholds through FastAPI and Streamlit.

JanSeva AI
Technologies: Amazon Bedrock, RAG, Transcribe, Polly, Translate, OpenSearch, AWS Lambda
- Built a voice-first multilingual RAG assistant that matches citizens to government welfare schemes, checks eligibility, generates pre-filled forms, and lists required documents.
- Integrated an Amazon Bedrock knowledge base covering 50+ schemes with Transcribe, Polly, Translate, OpenSearch, AWS Lambda, and low-bandwidth access paths.

F1 Race Position Predictor
Technologies: Python, Scikit-learn, XGBoost, Pandas, Gradio
- Engineered a gradient-boosting pipeline from 1,000+ races using driver form, qualifying, sprint, and circuit features; exposed predictions and MAE/R^2 evaluation through Gradio.

Multimodal Data Curation and Voice Summarization
Technologies: Python, Whisper, OpenCV, Google Vision, Translation APIs
- Built a multimodal data pipeline for audio, video, and images using Whisper transcription, OCR, translation APIs, normalized metadata, and automated validation.

ResearchHub
Technologies: React, TypeScript, PostgreSQL, Prisma, JWT, Zod
- Built a role-based research management application with JWT authentication, project membership, task assignment, progress dashboards, REST APIs, and PostgreSQL validation.

VanGuard Crowd Management System
Technologies: TypeScript, Predictive Analytics, Real-Time Monitoring
- Developed a pilgrimage-site crowd management prototype with live monitoring, virtual queue tokens, capacity tracking, and predictive congestion alerts.

Hackathons and Competitions

- **AWS AI for Bharat Hackathon: Finalist.** Built JanSeva AI, a multilingual voice and RAG assistant for government welfare access.
- **Data Science Conference Hackathon: Winner.** Built TRACE for AI-assisted grading, verification, analytics, and early identification of at-risk students.
- **SupeRR Selection Hackathon: Top 36 of 500+ participants.** Developed an IPL player-selection and bid-limit strategy based on team composition.
- **Agentathon:** Built Neural Consensus Engine with Gemini, LangGraph orchestration, and GCP deployment.
- **Smart India Hackathon:** Developed the VanGuard crowd monitoring, virtual queue, and predictive analytics prototype.

Education

University of Hyderabad (UoH) — **2024 - 2026**
M.Tech, Information Technology — *Hyderabad, India*
- Relevant coursework: Deep Learning, Machine Learning, Advanced Operating Systems, Social Network Analysis

Modern Education Society College of Engineering — **2019 - 2023**
B.E., Computer Engineering (Honors in AI/ML); CGPA: 8.85 — *Pune, India*